



**Western Australian Certificate of Education  
Examination, 2009  
Question/answer booklet**

**PHYSICAL  
EDUCATION  
STUDIES**

**Written paper  
Stage 2**

Please place your student identification label in this box

Student Number: In figures

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In words

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**Time allowed for this paper**

Reading time before commencing work: ten minutes  
Working time for paper: two and a half hours

**Materials required/recommended for this paper**

***To be provided by the supervisor***

This Question/Answer Booklet  
Multiple-Choice Answer Sheet

***To be provided by the candidate***

Standard items: pens, pencils, eraser, correction fluid, ruler, highlighters

Special items: non-programmable calculator

**Important note to candidates**

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

## Structure of this paper

This paper is worth 70 per cent of the total marks for the WACE examination in this course. The remaining 30 per cent of marks will come from the practical examination in this course.

| Section                        | Number of questions available | Number of questions to be attempted | Suggested working time (minutes) | Marks available |
|--------------------------------|-------------------------------|-------------------------------------|----------------------------------|-----------------|
| Section One: Multiple-Choice   | 20                            | 20                                  | 30                               | 20              |
| Section Two: Short Answer      | 10                            | 10                                  | 75                               | 50              |
| Section Three: Extended Answer | 4                             | 2                                   | 45                               | 30              |
| Total marks                    |                               |                                     |                                  | 100             |

## Instructions to candidates

1. The rules for the conduct of Western Australian external examinations are detailed in the *Year 12 Information Handbook 2009*. Sitting this examination implies that you agree to abide by these rules.
2. Section One: Answer **all** questions on the separate Multiple-Choice Answer Sheet. Use a blue or black pen or a B or 2B pencil.
3. Section Two: Answer **all** questions.
4. Section Three: Answer **two** of the four questions.
5. Write your answers to Section Two and Section Three in this Question/Answer Booklet.
6. Spare pages are provided at the back of this Question/Answer Booklet.

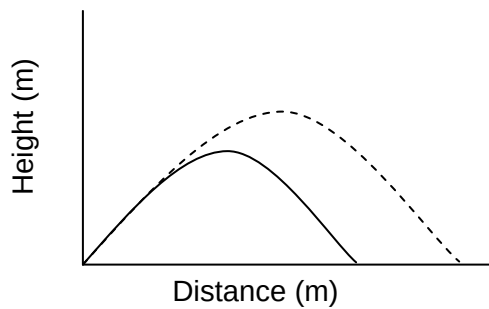
## Section One: Multiple-Choice 20 Marks

Attempt **all** questions in this section. Record your answers on the separate Multiple-Choice Answer Sheet using a blue or black pen or a B or 2B pencil. Each question is worth **one** mark.

Suggested working time for this section is 30 minutes.

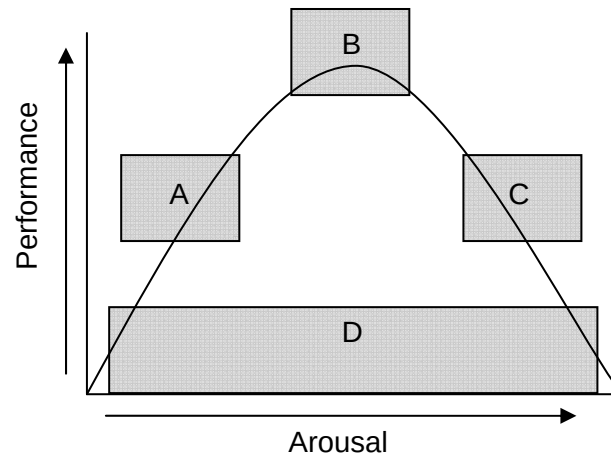
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1. Below is a diagram showing the flight path of two throws. Choose the answer that best explains why one throw travels a greater distance.



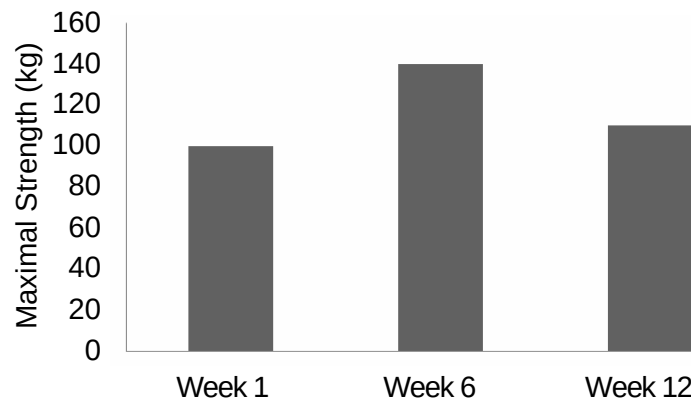
- (a) greater height of release
  - (b) greater angle of release
  - (c) greater velocity of projection
  - (d) greater inertia of the ball
2. After watching a videotape of yourself serving a tennis ball, you notice you are 'pulling up' on your follow through. What is the next phase of observation strategy you should implement?
- (a) observation
  - (b) diagnosis
  - (c) intervention
  - (d) re-observation

3. Which shaded section of the following diagram represents 'optimal' performance?



- (a) A  
(b) B  
(c) C  
(d) D
4. Before the game begins, your coach reviews the tactics and goals for the match. Which of the following communication skills is this coach utilising?
- (a) frontloading  
(b) debriefing  
(c) feedback  
(d) progressive overload
5. Which of the following best represents a positive method of stress management for a new professional athlete who has just had a poor opening match?
- (a) reading the papers the next day, allowing his self-confidence to fall  
(b) going out after the game and 'partying' to forget the match  
(c) writing down both the positives and negatives of his performance immediately after the match  
(d) lying down that night and replaying only the bad moments that occurred
6. A running coach notices an athlete is crossing his arms across the body, instead of along the body, which is considered 'wasted movement'. Based on this observation, the main focus for this athlete should involve improvement in
- (a) cardiovascular fitness.  
(b) muscle mass.  
(c) skill understanding.  
(d) movement efficiency.

7. After working on your free throw for several weeks, your coach expects you to be able to make seven out of every ten free throws you shoot in the game. This is an example of which category of transfer of learning?
- (a) training to competition
  - (b) skill to skill
  - (c) theory to practice
  - (d) execution to movement
8. If comparing two athletes, a sprinter and a long-distance runner, in which component of fitness would you expect the sprinter to excel over the endurance runner?
- (a) cardiorespiratory fitness
  - (b) muscular endurance
  - (c) balance
  - (d) muscular power
9. An athlete training for a forthcoming competition recorded his maximal strength levels at the start of his training (week 1), right before his competition (week 6) and after a break from training (week 12). These are shown on the graph below. Use this graph to determine which principle of training the athlete displayed from week 6 to week 12.



- (a) peaking
  - (b) progressive overload
  - (c) maintenance
  - (d) reversibility
10. Which of the following fuel sources will provide most of the ATP used during a rowing event that lasts two minutes?
- (a) fats
  - (b) carbohydrates
  - (c) protein
  - (d) creatine phosphate

11. Within two hours of an athletic event, participants should avoid which of the following fuel sources in a meal?
- (a) fats
  - (b) carbohydrates
  - (c) protein
  - (d) water
12. You have practised the block start of your swimming event for weeks. On the day of your race you notice that you are the first swimmer in the water. Which neuromuscular adaptation may be responsible for your improvement?
- (a) balance
  - (b) reaction time
  - (c) aerobic capacity
  - (d) increase of type I muscle fibres
13. During netball practice, one of the players sprained an ankle. In the weeks after the injury this athlete would undergo rehabilitation to
- (a) decrease range of motion.
  - (b) regain strength.
  - (c) decrease proprioception.
  - (d) increase inflammation.
14. The table below shows the adaptations from a person's training that occurred over 12 weeks. What method of training would they most likely have been undertaking?

| Adaptation            | Result    |
|-----------------------|-----------|
| Type I muscle fibre % | Increase  |
| Aerobic capacity      | Increase  |
| Capillarisation       | Increase  |
| Anaerobic capacity    | No change |

- (a) resistance training
- (b) continuous training
- (c) interval training
- (d) fartlek training

15. The image below shows two wrestlers 'locked' in position.

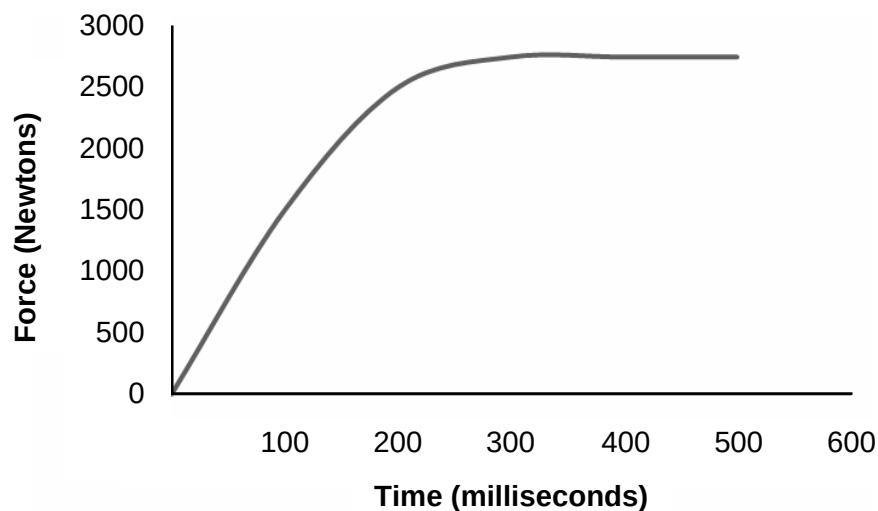
For copyright reasons this image cannot be reproduced in the online version of this document, but may be viewed at the link below:

<http://www.innfusion.com/blogimages/WrestlersSm.jpg>

The type of muscle contraction they are primarily using at this point is

- (a) isometric.
  - (b) isotonic.
  - (c) isokinetic.
  - (d) isotope.
16. A person's response to a 'find-thirty' advertising campaign on television was to start walking around the block regularly. This would place them in which stage of change?
- (a) pre-contemplation
  - (b) preparation
  - (c) action
  - (d) maintenance
17. While practising for the long jump on your own, you use the marks in the sand as a method of feedback for each attempt. What type of feedback are you using?
- (a) intrinsic
  - (b) concurrent
  - (c) verbal
  - (d) terminal
18. You are the instructor of several beginners' fitness classes. Which of the following strategies would you **avoid** to ensure maximum participation?
- (a) Provide the class at several different times.
  - (b) Provide multiple classes, catering to different levels of fitness.
  - (c) Provide exactly the same class at all times—'one class for all'.
  - (d) Provide a discount if participants enrol in a class for six consecutive weeks.

19. You are watching a bike race on the instant feedback computer. You notice that the acceleration of your favourite rider is a constant positive value during the sprint to the finish. The velocity feedback on the computer is broken, but you know the velocity will be
- (a) decreasing.
  - (b) remaining constant.
  - (c) increasing.
  - (d) the same value as the acceleration.
20. The graph below represents the force of a person pushing against a wall as hard as they can. According to this graph, which variable determines the maximal amount of force that can be applied?



- (a) impulse
- (b) inertia
- (c) velocity
- (d) time

**End of Section One**

## Section Two: Short Answer

50 Marks

Attempt all the following **ten (10)** questions.

Write your answers in the spaces provided.

Suggested working time for this section is 75 minutes.

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Question 1

(4 marks)

Susan has been training for the U/19 hockey team. She has noticed that the last four weeks of training have not yielded the improvements in cardiovascular fitness she had expected. Her fitness training included cycling and swimming for the same distances every week.

Name the **two** principles of training that Susan has neglected and explain what she might change with respect to these principles to improve her cardiovascular fitness.

Principle 1: \_\_\_\_\_

Explanation: \_\_\_\_\_

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Principle 2: \_\_\_\_\_

Explanation: \_\_\_\_\_

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**Question 2****(6 marks)**

John is learning how to perform a front kick in martial arts. Initially, he kicks with a stiff leg, as if his thigh, lower leg and foot are all one segment but, after some advice from a coach, John adapts his technique to look like the following:

For copyright reasons this image cannot be reproduced in the online version of this document.

- (a) A kick would be classified as what type of movement in the coordination continuum? (1 mark)
- 

- (b) Draw and label the graph below to demonstrate the coordination continuum of John's three segments during his high-velocity kick. (5 marks)



**Question 3****(4 marks)**

Motivation is an important aspect of sport performance. Name and define the two types of motivation that may influence an athlete and provide **one** example of each.

Motivation Type 1: \_\_\_\_\_

Explanation: \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

Motivation Type 2: \_\_\_\_\_

Explanation: \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

**Question 4****(6 marks)**

Goal setting is a strategy that may improve performance by focusing the athlete's attention, effort, motivation or persistence. There are several important considerations in setting effective goals.

Outline **three** considerations that affect goal setting, and provide an example of each factor. Examples can be from any sport.

Factor 1: \_\_\_\_\_

\_\_\_\_\_

Example: \_\_\_\_\_

\_\_\_\_\_

Factor 2: \_\_\_\_\_

\_\_\_\_\_

Example: \_\_\_\_\_

\_\_\_\_\_

Factor 3: \_\_\_\_\_

\_\_\_\_\_

Example: \_\_\_\_\_

\_\_\_\_\_

## Question 5

(6 marks)

The following figure shows the relative contribution of the three energy systems.

For copyright reasons this figure cannot be reproduced in the online version of this document.

- (a) Based on the graph, name the energy system that would be dominant for an event less than 10 seconds in length. (1 mark)

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- (b) Name and provide an example of an exercise or athletic event that would primarily stress the energy system labelled by line (A) in the diagram. (2 marks)

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- (c) A 400-metre race will last just under one minute. Describe the interplay of energy systems during the race. (3 marks)

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**Question 6****(6 marks)**

A sport coach is introducing the game of rugby to a group of novice players. Traditionally, the players would be taught 'touch' rugby, a non-contact version of the game, first. He avoids this tactic and instructs the players to just 'go at it'. In the first play, a player improperly tackles another, resulting in him falling to the ground and grabbing his knee in pain.

- (a) What did this coach neglect to consider when assigning this activity to the players?

(1 mark)

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- (b) The coach has a first aid certificate and applied the correct immediate care to the injured athlete. A doctor diagnosed him as having a knee sprain. The athlete is required to use crutches for next two weeks. What should the player's extended-care goals for rehabilitation be over the following three months, when he no longer requires crutches?

(5 marks)

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## Question 7

(6 marks)

Leadership styles have a direct influence on any group, especially in a team sport environment. Laissez-faire (casual) is an example of one style of leadership.

- (a) Name another style of leadership. (1 mark)

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- (b) Using the leadership style named in part (a), describe **three** advantages of this style of leadership. Answer in a sporting context. (3 marks)

1. 

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2. 

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3. 

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- (c) Outline **two** factors that influence the effectiveness of a chosen leadership style. (2 marks)

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2. 

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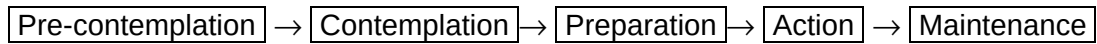
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## Question 8

(4 marks)

This question refers to the following model:



- (a) What is the name of the above behavioural model? (1 mark)

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Robyn is attempting to become more reliable as a member of the rowing crew. In Year 11, she missed a number of training sessions. She has developed a plan of action for Year 12.

- (b) At what phase of the model is Robyn? (1 mark)

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- (c) Half way through Year 12, Robyn has moved from the Action phase to the Maintenance phase. Describe **two** changes in her behaviour that would indicate she has made this change. (2 marks)

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## Question 9

(4 marks)

This question refers to the following diagrams.

For copyright reasons these diagrams cannot be reproduced in the online version of this document.

- (a) Which diagram (A or B) represents the push-off phase of the highest vertical jump? (1 mark)

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- (b) According to the information provided, outline **three** movement principles that increase jump height. (3 marks)

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## Question 10

(4 marks)

A softball pitcher is having difficulty with her pitching technique. She tells her coach that her ball release 'feels off'. Her coach tells her, 'All your pitches are missing the target. They are high by 40 centimetres'.

- (a) Name the type of feedback the player is providing. (1 mark)

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- (b) Name the type of feedback the coach is providing. (1 mark)

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- (c) There are two different forms of feedback a coach can use to analyse technique. Provide **one** example of each form to help the coach assist this softball pitcher. (2 marks)

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**End of Section Two**

## Section Three: Extended Answer

30 Marks

Attempt **two (2)** of the following four questions.

Write your answers in the spaces provided.

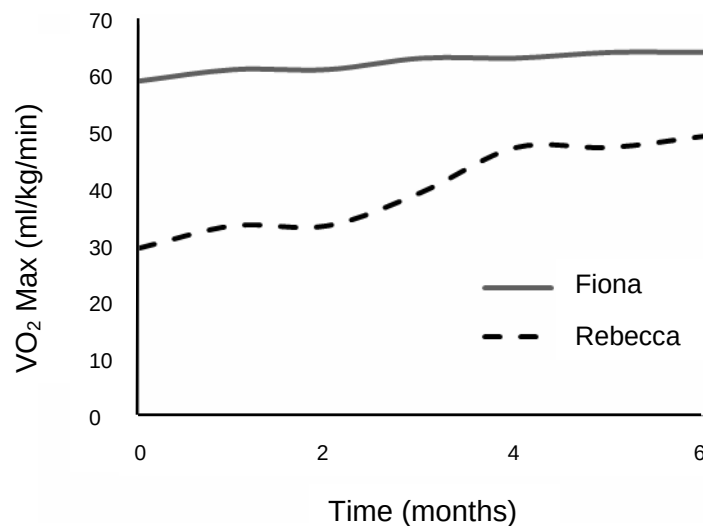
Suggested working time for this section is 45 minutes.

**Question 1****(15 marks)**

Fiona undertook a six-month training program prior to entering a marathon race. She had run marathons for many years and had a personal best time of 2 hours and 40 minutes. In the race, Fiona beat her personal best, finishing in 2 hours and 25 minutes.

Rebecca also undertook a six-month training program prior to entering the same marathon race. It was her first marathon. Previously she had maintained a healthy lifestyle by walking daily but had never exercised to a level where her breathing became rapid or difficult. Rebecca did well for her first race, finishing in 4 hours and 30 minutes.

Below is a graph of their  $\text{VO}_2$  max from the start of their training program until the day before the race.



- (a) Compare and explain why there is a difference between Fiona and Rebecca in the amount of improvement in their  $\text{VO}_2$  max. (5 marks)

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See next page

- (b) Other than a change in  $\text{VO}_2$  max, describe **ten** central and/or peripheral physiological responses to training you would expect to see in both Fiona and Rebecca after the six months of endurance training. (10 marks)

1. \_\_\_\_\_

\_\_\_\_\_

2. \_\_\_\_\_

\_\_\_\_\_

3. \_\_\_\_\_

\_\_\_\_\_

4. \_\_\_\_\_

\_\_\_\_\_

5. \_\_\_\_\_

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6. \_\_\_\_\_

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7. \_\_\_\_\_

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8. \_\_\_\_\_

\_\_\_\_\_

9. \_\_\_\_\_

\_\_\_\_\_

10. \_\_\_\_\_

\_\_\_\_\_

**Question 2****(15 marks)**

The Tour de France is a 21-stage cycling event. The cyclists can cover 200 km in a single stage, including extremely strenuous mountain climbs in which riders can reach near maximal heart rate levels. Most riders will have an average heart rate around 80 per cent of their maximum heart rate for up to six hours on the bike. During each stage cyclists are provided food by their support team or carry small amounts of food in their jersey.

- (a) For a single stage of the Tour de France, describe the nutritional strategies a cyclist should undertake before, during and after the event. Include in your description how each strategy will assist the athlete physiologically. (12 marks)

Before event: \_\_\_\_\_

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During the event: \_\_\_\_\_

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After event: \_\_\_\_\_

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- (b) Describe the physiological consequences that would result if a cyclist did not eat anything during the stage. (3 marks)

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### Question 3

**(15 marks)**

Chris is a gymnast who is trying to perform his first successful front flip (front somersault). This is a difficult movement in which a gymnast runs forward, jumps into the air and rotates forward 360 degrees before landing on his feet again. Below are sequential images of his first attempt.



Chris completed his first attempt but travelled too far during the flip and just barely made his rotation in time to land on his feet. His coach provided him with the following feedback for his second attempt: 'you need to run faster in your approach, increase your take-off angle and tuck tighter during your rotation'. Chris took in the information and realised that he hadn't run at full speed because he was nervous about his first attempt. His anxiety also caused him to be tight during his takeoff and rotation, not allowing his body to tuck comfortably.

- (a) Describe the biomechanical principles that explain why the corrections the coach has asked Chris to make will assist with his front flip performance. (9 marks)

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are approximately 20 lines visible. The paper has a slight shadow on its right side, suggesting it's resting on a surface.

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- (b) Name and provide an example of **three** mental skills strategies Chris could use to improve his performance. (6 marks)

Strategy 1: \_\_\_\_\_

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Strategy 2: \_\_\_\_\_

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Strategy 3: \_\_\_\_\_

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### Question 4

**(15 marks)**

You have been asked to teach a skill in the sport of your choice at a local school. You will be coaching a group of novices and a group of skilled players. Compare and contrast the coaching techniques and activities you could use to improve skill execution and increase skill adaptability for both groups. In your discussion, use specific examples of transfer of learning as well as coaching/training activities for enhanced performance.

[illegible]

[illegible]

**End of questions**

*Check that you have written your Student Number on the front cover of this booklet.*

Spare page: \_\_\_\_\_

[illegible]

Spare page: \_\_\_\_\_

[illegible]

## ACKNOWLEDGEMENTS

## Section One:

- Question 15** Innfusion Studio. (n.d.). *Wrestlers* [Image]. Retrieved November, 2009, from: <http://www.innfusion.com/blogimages/WrestlersSm.jpg>

## Section Two:

- Question 2** Adapted from: Physical Arts. (n.d.). *Front right kick* [Image]. Retrieved November, 2009, from: [http://www.physical-arts.com/training/techniques/mn\\_front\\_kick\\_right.html](http://www.physical-arts.com/training/techniques/mn_front_kick_right.html)
- Question 5** Image from: Brooks, G., Fahey, T., White, T., and Baldwin, K. (eds) (2000). *Exercise physiology: Human bioenergetics and its applications*. (3rd ed.). New York: McGraw-Hill, p. 29.
- Question 9** Image from: Hall, S.J. (2007). *Basic Biomechanics*. (5th ed.). New York: McGraw-Hill, p. 402.

## Section Three:

- Question 3** Vidic, M. (n.d.). *Front flip* [Image]. Retrieved November, 2009, from Flashmavi website: [http://www.flashmavi.com/gymnastics\\_floor\\_front\\_flip\\_how\\_to\\_do\\_a\\_front\\_sault.shtml](http://www.flashmavi.com/gymnastics_floor_front_flip_how_to_do_a_front_sault.shtml)

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